

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			26.6 (2) if answer incorrect award (1) for either of following $\frac{14.3}{53.7} / 0.266$ 0.266×100 no ecf possible alternative methods $\frac{14.3}{0.537}$ (1) $\frac{100}{53.7} \times 14.3$ (1) 26.6 (1) 26.6 (1) no ecf possible		2		2	2	
	(e)			a catalyst lowers the minimum energy needed for reaction / successful collisions / lowers the activation energy (1) this means more <u>successful</u> collisions (per second) / higher frequency/chance of <u>successful</u> collisions (so higher rate) (1) if no reference to collisions award (1) mark for 'a catalyst provides a suitable surface for a reaction'	2			2		
				Question 5 total	2	6	0	8	4	1